CURRICULUM STRUCTURE

SEMESTER – I

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20HSG01	Technical English	HS	2	0	2	4	3
2	20PHG01	Engineering Physics	BS	3	1	0	4	4
3	20CHG01	Engineering Chemistry	BS	3	1	0	4	4
4	20MHG01	Calculus and Linear Algebra	BS	3	1	0	4	4
5	20ITG01	Programming for problem solving using C	ES	3	0	0	3	3
PRACTICALS								
6	20PHG02	Engineering Physics Lab	BS	0	0	3	3	1.5
7	20MEG02	Engineering Workshop	ES	0	0	4	4	2
8	20ITG02	Programming in C Laboratory	ES	0	0	4	4	2
Total				**14**	**3**	**13**	**30**	**23.5**

SEMESTER – II

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20HSG02	Universal Human Values – II : Understanding Harmony	HS	3	0	0	3	3
2	20MHG02	Differential Equation and Complex Variables	BS	3	1	0	4	4
3	20CSG01	Object Oriented Programming using C++	ES	3	0	0	3	3
4	20CEG01	Engineering Mechanics	ES	3	0	0	3	3
5	20ME001	Manufacturing Processes I	PC	3	0	0	3	3
PRACTICALS								
6	20CHG02	Engineering Chemistry Lab	BS	0	0	3	3	1.5
7	20MEG01	Engineering Graphics	ES	0	0	4	4	2
8	20CSG02	Programming in C++ Laboratory	ES	0	0	4	4	2
9	20AC001	Environmental Science (AC-1)	AC	3	0	0	3	0
Total				**18**	**1**	**11**	**30**	**21.5**

SEMESTER – III

S.No.	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20MHG03	Transforms and Partial Differential Equations	BS	3	1	0	4	4
2	20ME002	Manufacturing Processes II	PC	3	0	0	3	3
3	20ME003	Mechanics of Materials	PC	3	0	0	3	3
4	20ME004	Fluid Mechanics and Machinery	PC	3	0	0	3	3
5	20ME005	Engineering Thermodynamics	PC	3	0	0	3	3
6	20XXEXX	Open Elective – I	OE	3	0	0	3	3
PRACTICALS								
7	20ME006	Manufacturing Technology Laboratory	PC	0	0	3	3	1.5
8	20ME007	Mechanics of Materials and Fluid Machinery Laboratory	PC	0	0	3	3	1.5
Total				**18**	**1**	**6**	**25**	**22**

SEMESTER – IV

S.No.	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20EEG03	Electrical and Electronics Engineering	ES	3	0	0	3	3
2	20ME008	Theory of Machines	PC	3	0	0	3	3
3	20ME009	Thermal Engineering	PC	3	0	0	3	3
4	20ME010	Engineering Materials and Metallurgy	PC	3	0	0	3	3
5	20MEPXX	Elective I (Professional)	PE	3	0	0	3	3
6	20XXEXX	Open Elective – II	OE	3	0	0	3	3
PRACTICALS								
7	20EEG04	Electrical Engineering Laboratory	ES	0	0	3	3	1.5
8	20ME011	Thermal Engineering Laboratory	PC	0	0	3	3	1.5
9	20ME012	Dynamics Laboratory	PC	0	0	3	3	1.5
Total				**18**	**0**	**9**	**27**	**22.5**

SEMESTER – V

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20ME013	CAD/CAM	PC	3	0	0	3	3
2	20ME014	Heat and Mass Transfer	PC	3	0	0	3	3
3	20ME015	Design of Machine Elements	PC	3	0	0	3	3
4	20ME016	Engineering Metrology and Quality Control	PC	3	0	0	3	3
5	20MEPXX	Elective II (Professional)	PE	3	0	0	3	3
6	20XXEXX	Open Elective – III	OE	3	0	0	3	3
PRACTICALS								
7	20ME017	CAD/CAM Laboratory	PC	0	0	3	3	1.5
8	20ME018	Heat and Mass Transfer Laboratory	PC	0	0	3	3	1.5
9	20ME019	Metrology Laboratory	PC	0	0	3	3	1.5
Total				**18**	**0**	**9**	**27**	**22.5**

SEMESTER – VI

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20ME020	Mechatronics	PC	3	0	0	3	3
2	20ME021	Design of Transmission Systems	PC	3	0	0	3	3
3	20ME022	Finite Element Analysis	PC	3	0	0	3	3
4	20MEPXX	Elective III (Professional)	PE	3	0	0	3	3
5	20XXEXX	Open Elective VI	OE	3	0	0	3	3
PRACTICALS								
6	20ME023	Mechatronics Laboratory	PC	0	0	3	3	1.5
7	20ME024	Simulation and Analysis Laboratory	PC	0	0	3	3	1.5
8	20ME901	Design Project	EC	0	0	8	8	4
9	20AC002	Constitution of India (AC-2)	AC	3	0	0	3	0
Total				18	0	14	32	22

SEMESTER – VII

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20HMG06	Professional Ethics in Engineering	HS	3	0	0	3	3
2	20ME025	Computer Integrated Manufacturing	PC	3	0	0	3	3
3	20MEPXX	Elective – IV (Professional)	PE	3	0	0	3	3
4	20MEPXX	Elective –V (Professional)	PE	3	0	0	3	3
PRACTICALS								
5	20ME902	Final Year Project Phase I	EC	0	0	6	6	3
Total				**12**	**0**	**6**	**18**	**15**

SEMESTER – VIII

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
THEORY								
1	20MEPXX	Elective – VI (Professional)	PE	3	0	0	3	3
PRACTICALS								
2	20ME903	Final Year Project Phase II	EC	0	0	16	16	8
Total				**3**	**0**	**16**	**19**	**11**

TOTAL NUMBER OF CREDITS:160

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PROFESSIONAL ELECTIVES

ELECTIVE – I [MATERIALS]

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	20MEP01	Composite Materials	PE	3	0	0	3	3
2	20MEP02	Non-destructive Evaluation of Materials	PE	3	0	0	3	3
3	20MEP03	Materials Characterization Techniques	PE	3	0	0	3	3
4	20MEP04	Nano Materials and Applications	PE	3	0	0	3	3
5	20MEP05	Metal Forming Technology	PE	3	0	0	3	3
6	20MEP06	Emerging Materials	PE	3	0	0	3	3

ELECTIVE – II [MANUFACTURING]

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	20MEP11	Unconventional Machining Processes	PE	3	0	0	3	3
2	20MEP12	Production Planning and Control	PE	3	0	0	3	3
3	20MEP13	Maintenance Engineering	PE	3	0	0	3	3
4	20MEP14	Additive Manufacturing	PE	3	0	0	3	3
5	20MEP15	Casting and Welding Processes	PE	3	0	0	3	3
6	20MEP16	Process Planning and Cost Estimation	PE	3	0	0	3	3

ELECTIVE – III [THERMAL]

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	20MEP21	Advanced Internal Combustion Engines	PE	3	0	0	3	3
2	20MEP22	New and Renewable Sources of Energy	PE	3	0	0	3	3
3	20MEP23	Gas Dynamics and Jet Propulsion	PE	3	0	0	3	3
4	20MEP24	Energy Conservation in Industries	PE	3	0	0	3	3
5	20MEP25	Refrigeration and Air Conditioning	PE	3	0	0	3	3
6	20MEP26	Automobile Engineering	PE	3	0	0	3	3

ELECTIVE – IV [DESIGN]

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	20MEP31	Product Design and Development	PE	3	0	0	3	3
2	20MEP32	Design for Manufacture & Assembly	PE	3	0	0	3	3
3	20MEP33	Piping System Design	PE	3	0	0	3	3
4	20MEP34	Computational Techniques For Fluid Dynamics	PE	3	0	0	3	3
5	20MEP35	Design Concepts in Engineering	PE	3	0	0	3	3
6	20MEP36	Design of Jigs, Fixtures and Press Tools	PE	3	0	0	3	3

ELECTIVE – V [MANAGEMENT]

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	20MEP41	Entrepreneurship Development	PE	3	0	0	3	3
2	20MEP42	Engineering Economics	PE	3	0	0	3	3
3	20MEP43	Lean Manufacturing	PE	3	0	0	3	3
4	20MEP44	Total Quality Management	PE	3	0	0	3	3
5	20MEP45	Operations Research	PE	3	0	0	3	3
6	20MEP46	Intellectual Property Rights	PE	3	0	0	3	3

ELECTIVE – VI [AUTOMATION]

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
1	20MEP51	Machine Vision	PE	3	0	0	3	3
2	20MEP52	Industrial Robotics Technology	PE	3	0	0	3	3
3	20MEP53	MEMS and Microsystems	PE	3	0	0	3	3
4	20MEP54	Applied Hydraulics and Pneumatics	PE	3	0	0	3	3
5	20MEP55	Industry 4.0	PE	3	0	0	3	3
6	20MEP56	Industrial Automation	PE	3	0	0	3	3

OPEN ELECTIVES

S.No	Course Code	Course Name	Category	Periods Per Week			Total Contact Periods	Credits
				L	T	P		
OFFERED BY COMPUTER SCIENCE ENGINEERING PROGRAMME								
1	20CSE01	Basics of Python Programming	OE	3	0	0	3	3
2	20CSE02	Introduction to AI	OE	3	0	0	3	3
3	20CSE03	Fundamentals of Data Science	OE	3	0	0	3	3
4	20CSE04	Basics of Internet Programming	OE	3	0	0	3	3
5	20CSE05	Introduction to Soft Computing	OE	3	0	0	3	3
OFFERED BY ELECTRONICS AND COMMUNICATION ENGINEERING PROGRAMME								
1	20ECE01	Electronic Measurements and Instrumentation	OE	3	0	0	3	3
2	20ECE02	Microcontrollers and its Applications	OE	3	0	0	3	3
3	20ECE03	Introduction to Embedded Systems	OE	3	0	0	3	3
4	20ECE04	Nano Electronics and Sensors	OE	3	0	0	3	3
5	20ECE05	Principles of VLSI Systems	OE	3	0	0	3	3
OFFERED BY INFORMATION TECHNOLOGY PROGRAMME								
1	20ITE01	Big Data Analytics and its Applications	OE	3	0	0	3	3
2	20ITE02	Cloud Computing Fundamentals	OE	3	0	0	3	3
3	20ITE03	Fundamentals of Internet of Things	OE	3	0	0	3	3
4	20ITE04	Introduction to Database Management Systems	OE	3	0	0	3	3
5	20ITE05	Web Interface Design and Development	OE	3	0	0	3	3
6	20ITE06	Introduction to Data Structures	OE	3	0	0	3	3

7	20ITE07	Principles of Software Engineering	OE	3	0	0	3	3
OFFERED BY ELECTRICAL AND ELECTRONICS ENGINEERING PROGRAMME								
1	20EEE01	Energy Management Systems	OE	3	0	0	3	3
2	20EEE02	Medical Instrumentation	OE	3	0	0	3	3
3	20EEE03	PLC Programming	OE	3	0	0	3	3
4	20EEE04	Renewable Energy Systems	OE	3	0	0	3	3
5	20EEE05	Virtual Instrumentation & Data Acquisition	OE	3	0	0	3	3
6	20EEE06	Electric Vehicles	OE	3	0	0	3	3

OPEN ELECTIVES OFFERED BY DEPARTMENT OF MECHANICAL ENGINEERING

Sl. No.	Course Code	Course Title	Category	Credit			Total Contact Periods	Credits
				L	T	P		
1	20MEE01	Automotive Fundamentals	GE	3	0	0	3	3
2	20MEE02	Computer Aided Design	GE	3	0	0	3	3
3	20MEE03	Introduction to Power Plant Engineering	GE	3	0	0	3	3
4	20MEE04	Introduction to Robotics	GE	3	0	0	3	3
5	20MEE005	3D Printing	GE	3	0	0	3	3

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